

Building Classification Using a YOLO Object Detection Model Trained on Aerial Imagery

Connor Williams*, Gregory Macfarlane*, Kaylen Turley*

* Department of Civil and Construction Engineering

Brigham Young University, Provo, Utah 84604

gregmacfarlane@byu.edu, willicon@byu.edu, kaylentu@byu.edu

Abstract—Travel forecasting models rely heavily on land use data from government agencies, which can be inconsistent or unavailable in many regions. In this paper we investigate the viability of using aerial imagery and machine learning models as a first step to develop a method to obtain this statistical data independently from government agencies. Using Logan, Utah as a primary case study, we used the Microsoft Building Footprints dataset and aerial imagery to train a “You Only Look Once” (YOLO) model to classify building types, including residential, commercial, and apartment buildings. We found that the model demonstrated high accuracy in identifying building types in the training city. However, when validated against five other Utah cities selected for diversity of median income and housing ages, the model showed limited transferability, which suggests that differences between house architecture could necessitate local training. We analyzed these performance discrepancies and discuss future potential research such as integrating additional environmental indicators and using high-resolution drone imagery to create more robust models for estimating household statistics and land use data.

Index Terms—aerial imagery; AI; land use data; machine learning; transportation

I. INTRODUCTION

Travel forecasting models rely on land use data as a primary input. These data include the number of persons and households living in a given area, as well as other information relevant to travel behavior including the availability of motor vehicles, the income of the residents, and the presence of workers, students, and non-workers. In many contexts, these data are assembled from government statistical agencies; in the United States common sources include the US Census Bureau and the Bureau of Labor Statistics [1]. In countries where data are available, these models are highly reliant on the consistent and public release of the data [2]. Previous studies have shown that these models suffer from a slight over-prediction bias in predicted traffic [3]. In countries where statistical agencies are not reliable, travel forecasting models struggle to be developed or have data gaps [4]. Standard practice has been to employ model transferability, where a model developed in one region is applied to another. This method works when the two regions have similar characteristics; however, is limited otherwise [5, 6].

Recent research has sought to find alternative methods to collect land use characteristics. Satellite aerial imagery has been used to analyze the volume and density of man-made

structures and roads to estimate area populations, in both urban and rural settings [7, 8]. Daytime imagery has also been used to track vehicle movement between major cities as an indication of economic activity [9]. Such imagery has also been used to classify land use by estimating landscape patterns and building functions, identifying different surfaces, and then categorizing building types [10]. Nighttime light data has also been shown to be useful as well. One study correlated city light output with socioeconomic variables such as population, GDP, and electric power consumption [11], while another used the light emitted from cars at night on highways to predict vehicle travel between cities [12]. Satellite imagery has also been used to produce datasets such as the Microsoft Building Footprints dataset, which has been used along with LiDAR and point of interest data to inventory residential structures from census data [13, 14]. Unfortunately, processes and methods reliant on satellite data are also susceptible to the availability of said data, which, like statistical data, may not be available across the world or could have its public access revoked.

More recent advances in machine learning have led to more applications of remote sensing data, including tracking vehicle movement between cities [9], or predicting population characteristics and locations of man-made structures across entire countries [8]. Models such as Ultralytics’ “You Only Look Once” (YOLO) model [15] are being used to identify buildings and extract building footprints [16, 17], a particularly useful application for travel forecasting model inputs. It is expected that as travel forecast and other datasets continue to grow larger the use of machine learning models for analysis will become increasingly relevant and practical.

Within the context of this background and research, the question arises as to whether and to what extent remote sensing data can be used to develop land use independently of statistical agencies. The development of a process or method that could synthesize such data would solve many problems in places without reliable statistical data as well as supplement data in other places.

In this paper, we present initial efforts to determine building type from building footprint data and aerial images of buildings. We do this by training a deep learning model on labeled data from Logan, Utah. We then apply this model to other cities in Utah to assess its viability in predicting building categories outside of the training data. The paper closes by

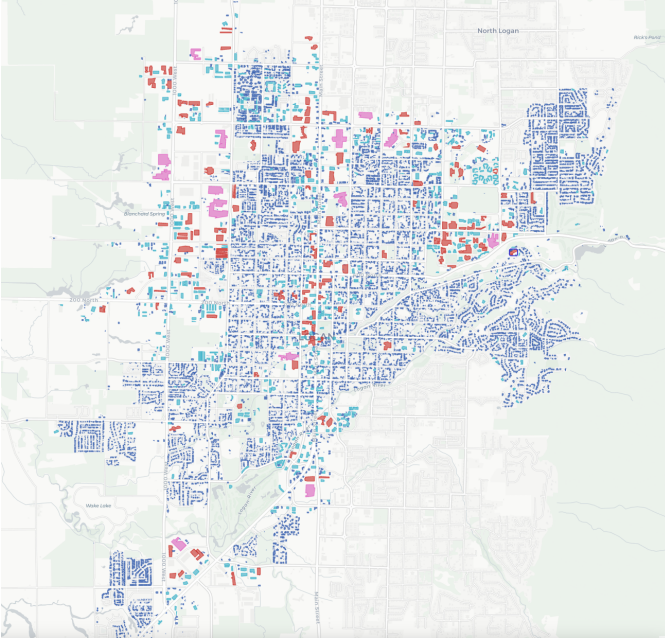


Figure 1: K-mean analysis for Logan, Utah

discussing limitations and associated future research.

II. METHODOLOGY

A. Data Collection

We chose the city of Logan, Utah, for the initial analysis due to it being a smaller city and having a variety of building ages and types. Using the latitude and longitude of the official city boundaries, all of the building footprints in the city were extracted from the Microsoft Building Footprints dataset. We chose to use this dataset because it allowed for easy access to building footprint data for this initial study. After calculating the area and perimeters of the individual footprint data, we ran a K-mean analysis to categorize differently sized buildings into distinct clusters. Figure 1 shows the distribution of cluster types, with the majority of structures being small residential buildings.

From each of these clusters, we obtained aerial images for a sample of 600 buildings in Logan using the Python package Contextily [18], an open-source program that retrieves satellite images from the Bing aerial image library, using bounding boxes to obtain small areas of the image tiles. Because the majority of the buildings were in the smallest cluster, we oversampled 200 buildings from clusters that included only larger buildings.

B. Model Training

Ultralytics’ “You Only Look Once”, or YOLO model, is an open-source object detection model that has become very popular within the research community due to its accuracy and ease of use [15]. The model is trained on images that have been manually annotated with bounding boxes to identify specific categories desired for the model to detect. In preparation for



Figure 2: YOLO prediction output example

training the YOLO model, we annotated the collected satellite images using the Python package Label Studio [19]. Label Studio is an open-source tool that makes it easy to manually label images for use in machine learning. Student researchers manually labeled the images, sorting the buildings into five categories: Apartment, Residential, Commercial, Church, and School. A random sample of 80% of the images was selected for model training, with the other 20% of the images reserved for model validation. Within these image sets, clusters that included larger and less common building sizes were specifically included, due to their relatively lower likelihood of appearing, to make sure the model would be trained on these less common building types. Despite this, building types such as schools and church buildings were still underrepresented and appeared only a few times, due to their relatively low number even within larger buildings.

The YOLO model is trained on the training images over hundreds of iterations, until it can consistently recognize the different categories in the pictures. The YOLO model then validates itself on the holdout sample of annotated images not included in the training set to ensure the accuracy of training. After the validation, the model is then able to predict categories in nonannotated images by bounding them in bounding boxes and displaying the probability of correct identification. Figure 2 shows an example of a YOLO model prediction output for a sample of buildings in the dataset. The results of the validation can be seen in Figure 3.

C. Model Validation and Comparison

After this training, we tested the model’s predictive capabilities on other cities in Utah to assess the model’s viability in predicting building categories outside of the training sample. We chose five test cities based on similarities and dissimilarities in median income and age of housing stock as recorded American Community Survey data, shown in Table Ia. We theorized that these factors would have a significant impact on

Table I

(a) Chosen Cities by Median Income and House Age

House Age	High Income	Middle Income	Low Income
Old Houses	Bountiful, Utah	Ogden, Utah	Price, Utah
New Houses	Saratoga Springs	St. George, Utah	(None)

the architecture of the house and therefore the YOLO model’s ability to recognize and categorize them.

Following the same method as described above, we collected and clustered building footprint data for each city, randomly selecting and labeling 200 satellite images of buildings across building clusters. These annotated images were then run through the model using the validation modules only.

III. RESULTS AND DISCUSSION

A. Results From Logan, Utah

The results of the Logan trained YOLO model are shown in Figure 3. The desired output of the model is to have all the numbers in a diagonal line, indicating perfect prediction of the building’s category in line with the annotated labels. The confusion matrix shows that the model accurately predicted residential buildings 99% of the time, and all commercial and apartment buildings. However, it did underperform with churches (67%) due to the low number of buildings to train on.

An important aspect of the model to discuss is when it predicts different building types where there is no building. This is seen on the far right of each graph, where “background” is given as the true value, but the model gave predictions from every category. This error comes from two sources. The first is due to how we made the annotations in the training of the model. Auxiliary buildings, such as backyard sheds, were not consistently identified and annotated, but the model still recognized them as buildings and categorized them. There were also times when only a small portion of a building was present in an image, and we believed it to be too little of the building for the model to pick up, and so we did not annotate it. However, the model in some cases still recognized and categorized the building at a probability level, while in other cases it did not recognize partial buildings at all.

The other source of error was from the quality of the photos used. The satellite images from Contextily were convenient and easy to access but were often low quality. This resulted in the model sometimes interpreting objects of solid colors, such as roads and covered parking structures, as a type of building. Between these errors, the second one could be solved using higher quality photos to help the model better distinguish between these surfaces. The first error could be improved by implementing better guidelines on the annotation process to ensure consistency between datasets. However, in practice, the first error may be less relevant if only the primary building in each image matters to be identified.

B. Results From Other Utah Cities

When the Logan model was applied to other cities in Utah, it produced mixed results, as seen in Figure 4. In every city the model struggled to correctly distinguish between the different building types. Most of the building images we annotated for these cities were residential housing; however, the model categorized residential buildings incorrectly 50% of the time, if not more. Commercial and apartment buildings displayed worse results on average, with correct identification of 19% to 60% and 0% to 50%, respectively. Church and school buildings, if they appeared in the datasets, were misidentified 100% of the time. While it seemed to do better in Saratoga Springs with residential buildings, its accuracy with other building types is still far too low to be useful. It remains unclear why Saratoga Springs’ residential buildings were so well identified, with the city sharing little with Logan in respect to median income, house age, and location.

The model also struggled with simply recognizing buildings at all. Each city had a significant number of buildings that were predicted as “background”, meaning the model did not recognize them as buildings despite them being annotated. When we checked the building images, none of them had any characteristics, such as large trees or shadows partially covering the roof, that the Logan images did not also have. The quality of the images from Contextily are also likely not a factor, since they were the same across all the cities in Utah.

While we thought that the median income or house age could also play a significant factor in the model’s ability to recognize different building types, the results of the validations did not support this. The model performed roughly the same in every city it was not trained on. Neither house age nor median income made a difference in the model’s ability to classify. Further research is needed to identify what factors caused the model to perform so poorly on these cities.

Given the results of the validations, we would conclude that developing a process for model creation would be more useful and viable than creating a universal model to be used in a region. It appears that building styles and types are unique enough between cities that a single unified model, at least one mostly dependent on identifying building roofs, is not feasible. However, as we discuss more in the future research section, adding other environmental variables to the classification, such as the presence of sidewalks, swing sets, etc. could allow for a more general model to be developed.

IV. CONCLUSION AND FUTURE RESEARCH

In this paper, we presented the results of using a machine learning model to classify building types from aerial imagery. We used building footprints in Logan, Utah, to collect building images and annotated them to train a YOLO model. That model was then tested on predicting building categories from other cities in Utah, in which it was not trained. We found that the model did well in identifying buildings in the city it was trained in but performed poorly when it tried to predict buildings in other cities in Utah.

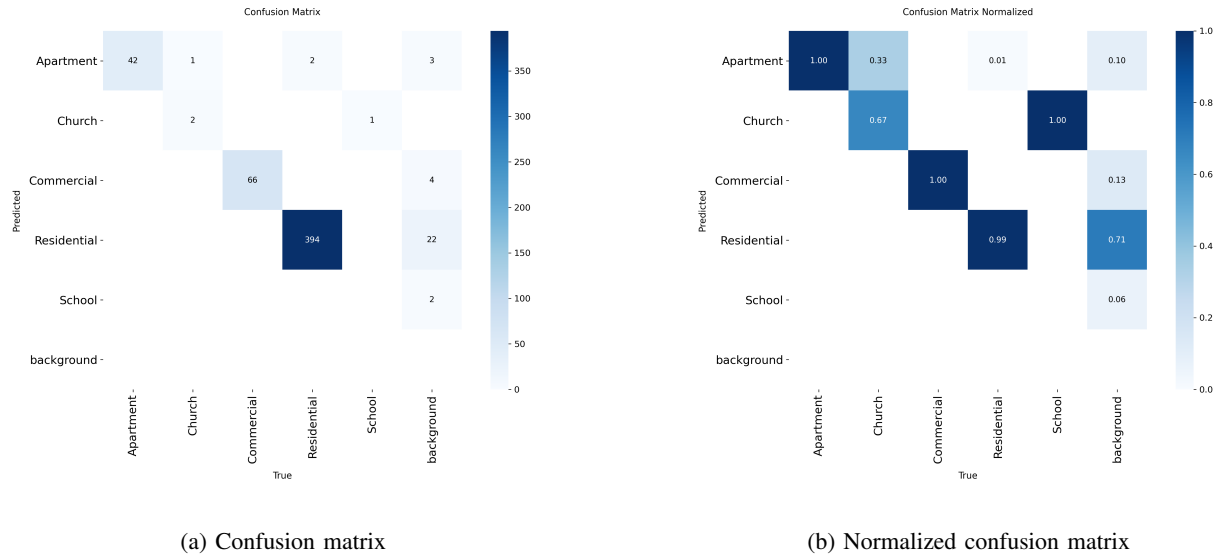


Figure 3: YOLO confusion matrices for Logan, Utah.

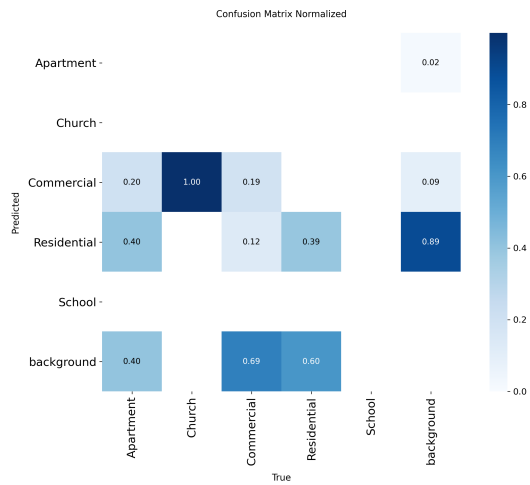
To make the model more accurate in categorizing building types, there are several different directions future research could take. One approach would be to improve the current model’s performance. This could be achieved using higher quality aerial imagery that we obtain from using drones. The rise of commercially available and affordable aerial drones has improved the ability for individuals and small organizations to capture high-quality aerial imagery, with impacts felt across many fields [20, 21]. Drone-acquired imagery could allow the model to distinguish houses on a much finer level, which could help with the model’s transferability. Another primary advantage of aerial drones vis-à-vis satellite imagery is the ability to capture images from multiple angles, which can be used to create a more complete picture of a given area [22]. Part of this more complete picture could be extracting better building footprint data. While the Microsoft Building Footprints dataset is expansive, it is not exhaustive. It is several years old and lacks footprints for recent developments. It also features many notable errors in building footprint identification, such as combining buildings together or creating obviously incorrect footprints. Creating our own process to extract footprints over a small area could improve the accuracy of our footprints and help not just in finding buildings but also in training other models.

Another direction future research could take would be to create a general model that would not only categorize a building but also be able to predict socioeconomic data for the building’s household. This model would incorporate many variables, with the YOLO model’s output being only one of them. Other variables could include the presence of objects that suggest socioeconomic status, such as vehicles, swing sets, and foliage around a house. These variables would come from other YOLO models trained specifically to identify these objects. Other inputs for the model could come from other types of remote sensing data. Data such as thermal radiation,

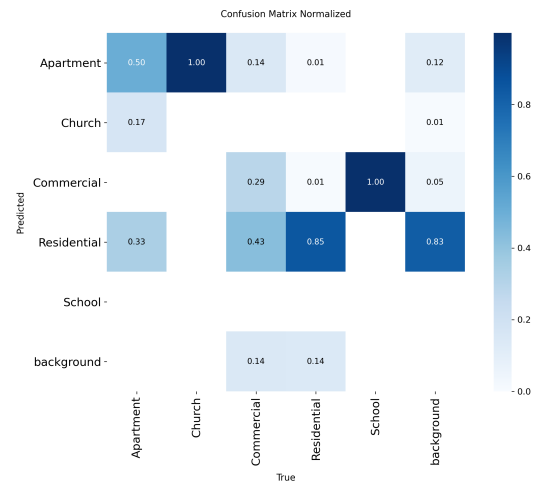
nighttime light data, or vegetation index data could be used to predict statistics like income calculated from the household’s heat use, light use, or amount of vegetation on a property. This data, from heat use to the presence of cars, could also give insights on the number of people living within a household, or whether children are present. There are potential ethical problems with this data collection, with drone-collected imagery challenging the boundary between public and private information [23]. More research is needed to address these potential privacy concerns

The analysis of the classification of buildings near each other is another variable that could be useful in a general model. Similar building types are often built next to each other. Thus, a building’s classification could be influenced by the predicted classification of the building around it. This would help in situations where a building may look residential, but is surrounded by warehouses, meaning it is likely some type of commercial support building. Variables based on nearby buildings could, however, run into issues in areas where mixed-use housing is common, or where residential and commercial buildings are more intertwined.

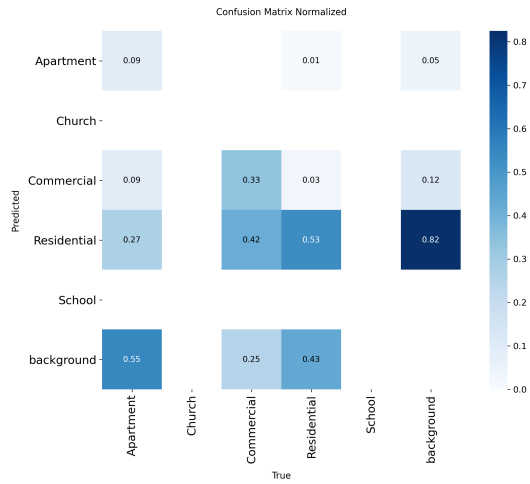
Currently, this research is susceptible to many of the same limitations as transportation models. Satellite and aerial imagery are not consistent around the world, having varying availability and image quality. This imagery may also be outdated, especially in fast developing areas. The use of commercial drones could solve these problems, but they also have challenges of their own. Collecting a whole city’s worth of imagery would take time to plan logistics and photograph. The flights would also be dependent on the weather, as well as local and national regulations that could present challenges. Splitting neighborhoods into subsections for data collecting and analysis could help mitigate these issues, though further research into laws and regulations would need to be conducted.



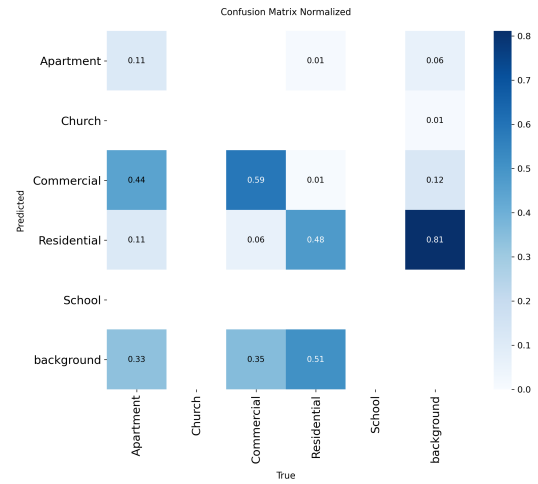
(a) Bountiful, Utah



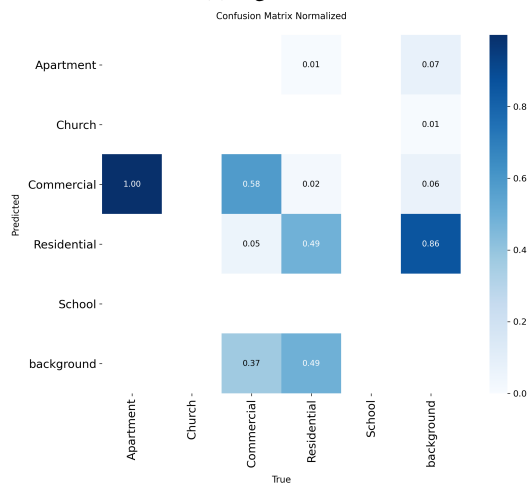
(b) Saratoga Springs, Utah



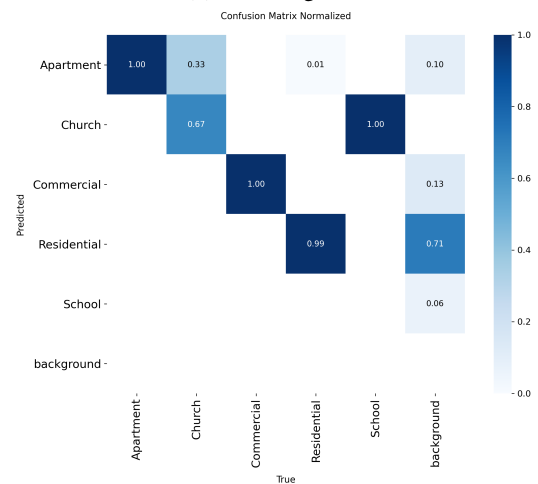
(c) Ogden, Utah



(d) St. George, Utah



(e) Price, Utah



(f) Logan, Utah

Figure 4: Normalized YOLO confusion matrices

REFERENCES

- [1] Inc. Cambridge Systematics et al. "Travel Demand Forecasting: Parameters and Techniques". In: *NCHRP Report 716*. Washington, D.C: Transportation Research Board, 2012, pp. 55–60.
- [2] Kathleen Rohanna and Jeff Tayman. "Census data for transportation planning, analysis, and implementation". In: *Journal of Economic and Social Measurement* 31.3–4 (Aug. 2006), pp. 167–183. DOI: 10.3233/JEM-2006-0276.
- [3] G. D. Erhardt et al. *Traffic Forecasting Accuracy Assessment Research*. Washington, D.C: Transportation Research Board, 2020, pp. II–44, III–C–3. DOI: 10.17226/25637.
- [4] Sameer Abu–Eisheh, Mohammed S. Ghanim, and Anas Dodeen. "Trip generation model for a developing city in an emerging country". In: *Transportation Research Interdisciplinary Perspectives* 24 (Mar. 2024), p. 101048. DOI: 10.1016/j.trip.2024.101048.
- [5] C. G. Wilmot and P. R. Stopher. "Transferability of Transportation Planning Data". In: *Transportation Research Record: Journal of the Transportation Research Board* 1768.1 (Jan. 2001), pp. 36–43. DOI: 10.3141/1768-05.
- [6] J. L. Bowman and M. Bradley. "Testing Spatial Transferability of Activity-Based Travel Forecasting Models". In: *Transportation Research Record: Journal of the Transportation Research Board* 2669.1 (Jan. 2017), pp. 62–71. DOI: 10.3141/2669-07.
- [7] Kamel Allaw et al. "Population estimation using geographic information system and remote sensing for unorganized areas". In: *Geoplanning: Journal of Geomatics and Planning* 7.2 (Nov. 1, 2020), pp. 75–86.
- [8] Casper Samsø Fibæk, Carsten Keßler, and Jamal Joker Arsanjani. "A multi-sensor approach for characterising human-made structures by estimating area, volume and population based on sentinel data and deep learning". In: *International Journal of Applied Earth Observation and Geoinformation* 105 (Dec. 25, 2021), p. 102628.
- [9] Patrick Lehnert et al. "Proxying economic activity with daytime satellite imagery: filling data gaps across time and space". In: *PNAS Nexus* 2.4 (Apr. 1, 2023), pgad099.
- [10] Ye Zhang et al. "Landscape patterns and building functions for urban land-use classification from remote sensing images at the block level: A case study of wuchang district, wuhan, china". In: *Remote Sensing* 12.11 (Jan. 2020), p. 1831.
- [11] Shankar Acharya Kamarajugedda, Pradeep V Mandapaka, and Edmond Y.M Lo. "Assessing urban growth dynamics of major southeast asian cities using nighttime light data". In: *International Journal of Remote Sensing* 38.21 (Nov. 2, 2017), pp. 6073–6093.
- [12] Ying Chang et al. "A novel method of evaluating highway traffic prosperity based on nighttime light remote sensing". In: *Remote Sensing* 12.1 (Jan. 2020), p. 102.
- [13] Microsoft. *GlobalMLBuildingFootprints*. Version Python. Jan. 7, 2026. URL: <https://github.com/microsoft/GlobalMLBuildingFootprints> (visited on 01/07/2026).
- [14] Xinyue Ye et al. "Enhancing population data granularity: A comprehensive approach using lidar, POI, and quadratic programming". In: *Cities* 152 (Sept. 2024), p. 105223.
- [15] Glenn Jocher, Jing Qiu, and Ayush Chaurasia. *Ultralytics YOLO*. Version 8.0.0. Jan. 2023. URL: <https://github.com/ultralytics/ultralytics>.
- [16] Wahidya Nurkarim and Arie Wahyu Wijayanto. "Building footprint extraction and counting on very high-resolution satellite imagery using object detection deep learning framework". In: *Earth Science Informatics* 16.1 (Mar. 1, 2023), pp. 515–532.
- [17] Mayank Sharma and Rahul Dev Garg. "Building footprint extraction from aerial photogrammetric point cloud data using its geometric features". In: *Journal of Building Engineering* 76 (Oct. 1, 2023), p. 107387.
- [18] Dani Arribas-Bel. *contextily: context geo tiles in Python*. 2023. URL: <https://contextily.readthedocs.io>.
- [19] Maxim Tkachenko et al. *Label Studio: Data labeling software*. Open source software available from <https://github.com/HumanSignal/label-studio>. 2020–2025. URL: <https://github.com/HumanSignal/label-studio>.
- [20] I. Colomina and P. Molina. "Unmanned aerial systems for photogrammetry and remote sensing: A review". In: *ISPRS Journal of Photogrammetry and Remote Sensing* 92 (June 2014), pp. 79–97. DOI: 10.1016/j.isprsjprs.2014.02.013.
- [21] Adam C. Watts, Vincent G. Ambrosia, and Everett A. Hinkley. "Unmanned Aircraft Systems in Remote Sensing and Scientific Research: Classification and Considerations of Use". In: *Remote Sensing* 4.6 (June 2012), pp. 1671–1692. DOI: 10.3390/rs4061671.
- [22] Bryce E. Berrett et al. "Large-Scale Reality Modeling of a University Campus Using Combined UAV and Terrestrial Photogrammetry for Historical Preservation and Practical Use". In: *Drones* 5.4 (Dec. 2021), p. 136. ISSN: 2504-446X. DOI: 10.3390/drones5040136. (Visited on 01/20/2026).
- [23] Ola Hall and Ibrahim Wahab. "The use of drones in the spatial social sciences". In: *Drones* 5.4 (Dec. 2021), p. 112.